

SWARMPLOT

A swarmplot is a type of plot used to show the distribution of individual data points across categories.

Each dot represents one data point.

The dots are spread out (like a swarm of bees) so that they don't overlap, making it easy to see how many points are in each category.

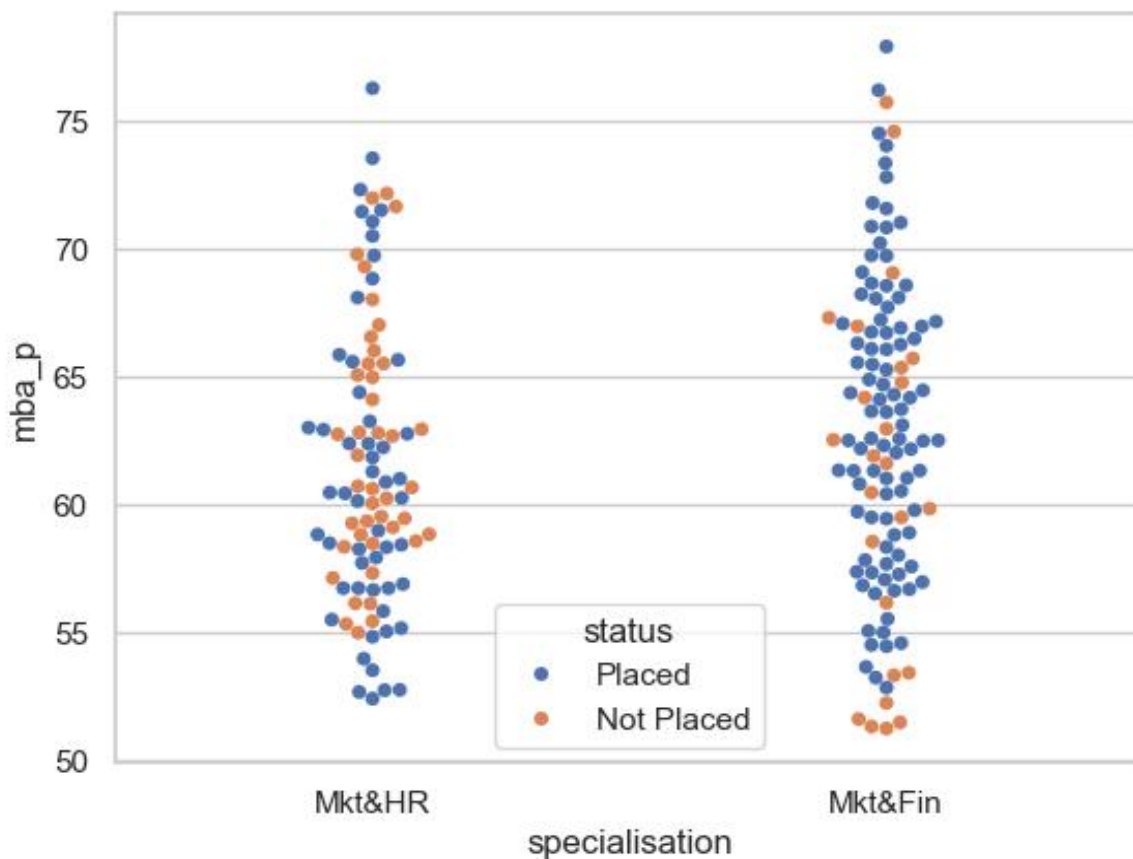
It is useful when you want to see both the overall distribution **and** each individual observation, instead of just a summary (like a boxplot).

For eg: This plot shows the MBA Percentage marks taken by individual students, on different Specialisations, from which how many of them are Placed and Not Placed.

From the Plot, we understand :

1. Most students prefer Mkt & Fin instead of Mkt & HR
2. Students who did Mkt & HR have scored less when compared to Mkt & Fin Specialization.
3. Many of students who did Mkt & HR are not Placed when compared to Mkt & Fin Specialization.

```
sns.swarmplot(data, x='specialisation', y='mba_p', hue='status')
```



VIOLINPLOT

A **violin plot** is a graph that shows the **distribution of data** across categories.

It looks like a **violin shape** because it combines a **boxplot** (showing median and spread) with a **density plot** (showing the shape of the distribution).

It helps to see both **summary statistics** (like a boxplot) and **the full distribution** (like a histogram or KDE).

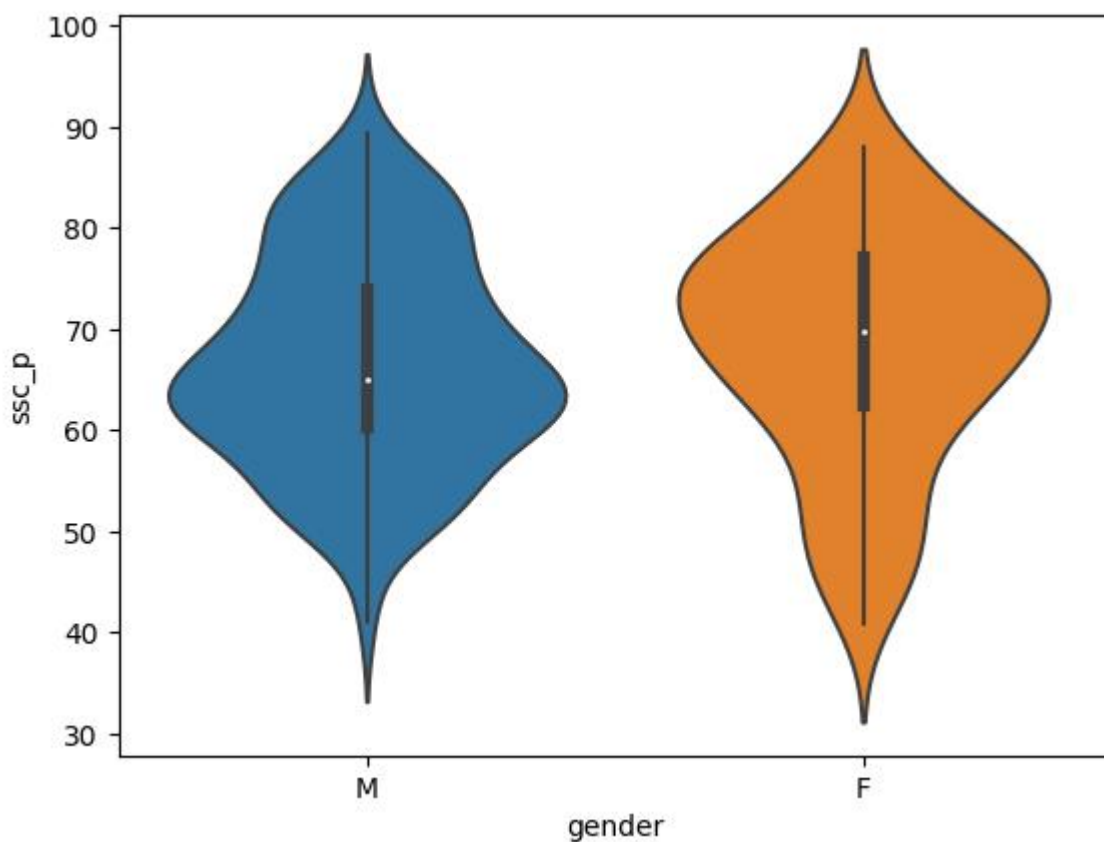
Here are some **observations from the violin plot** (ssc_p vs gender):

The distribution of ssc_p is more spread out, indicating varied performance across the range for Male, whereas Female distribution appears slightly more compact around higher scores, suggesting more consistency.

The median is higher for Females than Males, indicating a slightly better median performance.

The IQR is slightly higher for Females, meaning their middle 50% of scores are generally higher compared to Males.

Males have a slightly broader lower tail, indicating more low-percentage scores compared to Females.

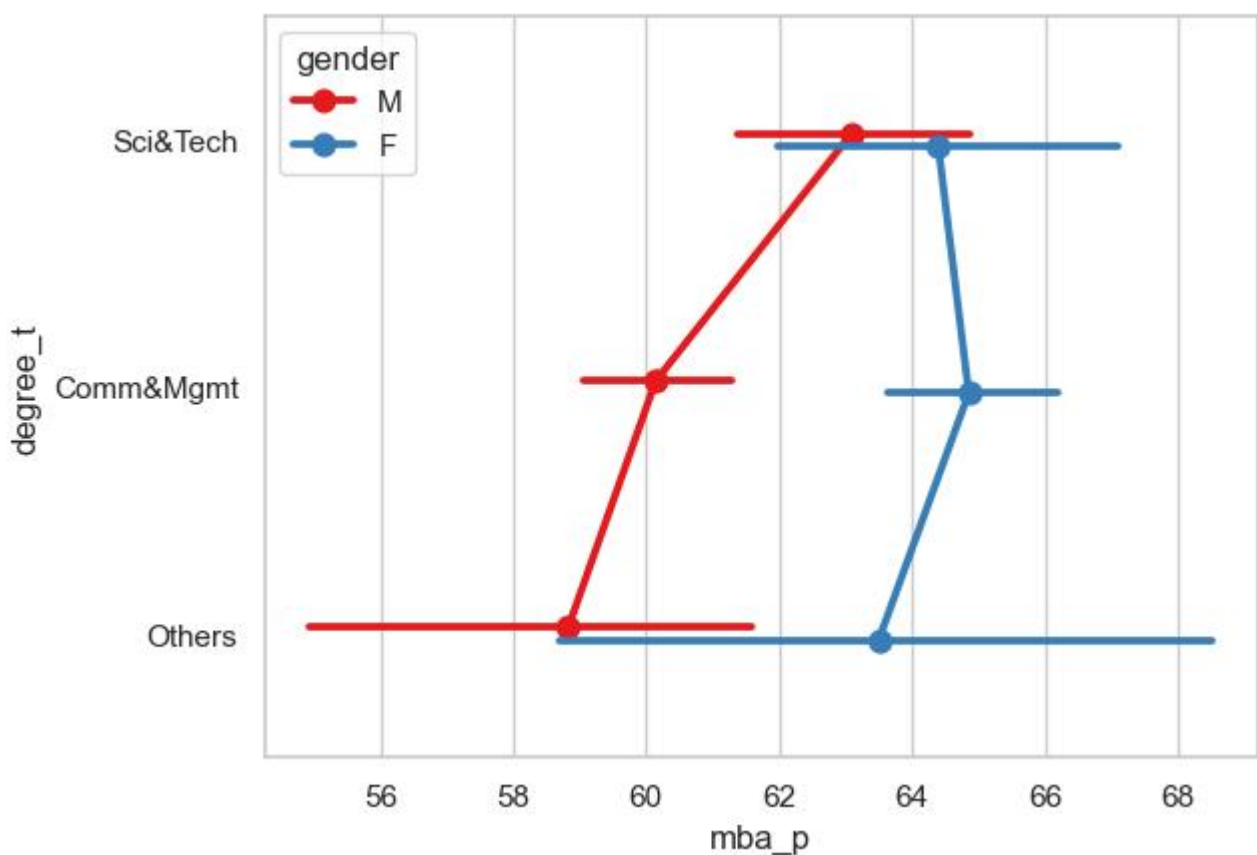


Point Plot

A **Point plot** represents an estimate of central tendency for a numeric variable by the position of the dot and provides some indication of the uncertainty around that estimate using error bars.

Point plots can be more useful than bar plots for focusing comparisons between different levels of one or more categorical variables. They are particularly adept at showing interactions: how the relationship between levels of one categorical variable changes across levels of a second categorical variable.

```
sns.pointplot(data, x='mba_p', y='degree_t', hue='gender', dodge=True)  
plt.show()
```



Here are some **Insights** from Point Plot for MBA Percent Based on various categories like Degree Majors and Gender:

Sci&Tech: Gender differences for MBA Percent is negligible.

Comm&Mgmt: Females outperform slightly.

Others: Strong gender gap, females well ahead.

Overall: Females have higher MBA percentages than males in most degree categories.

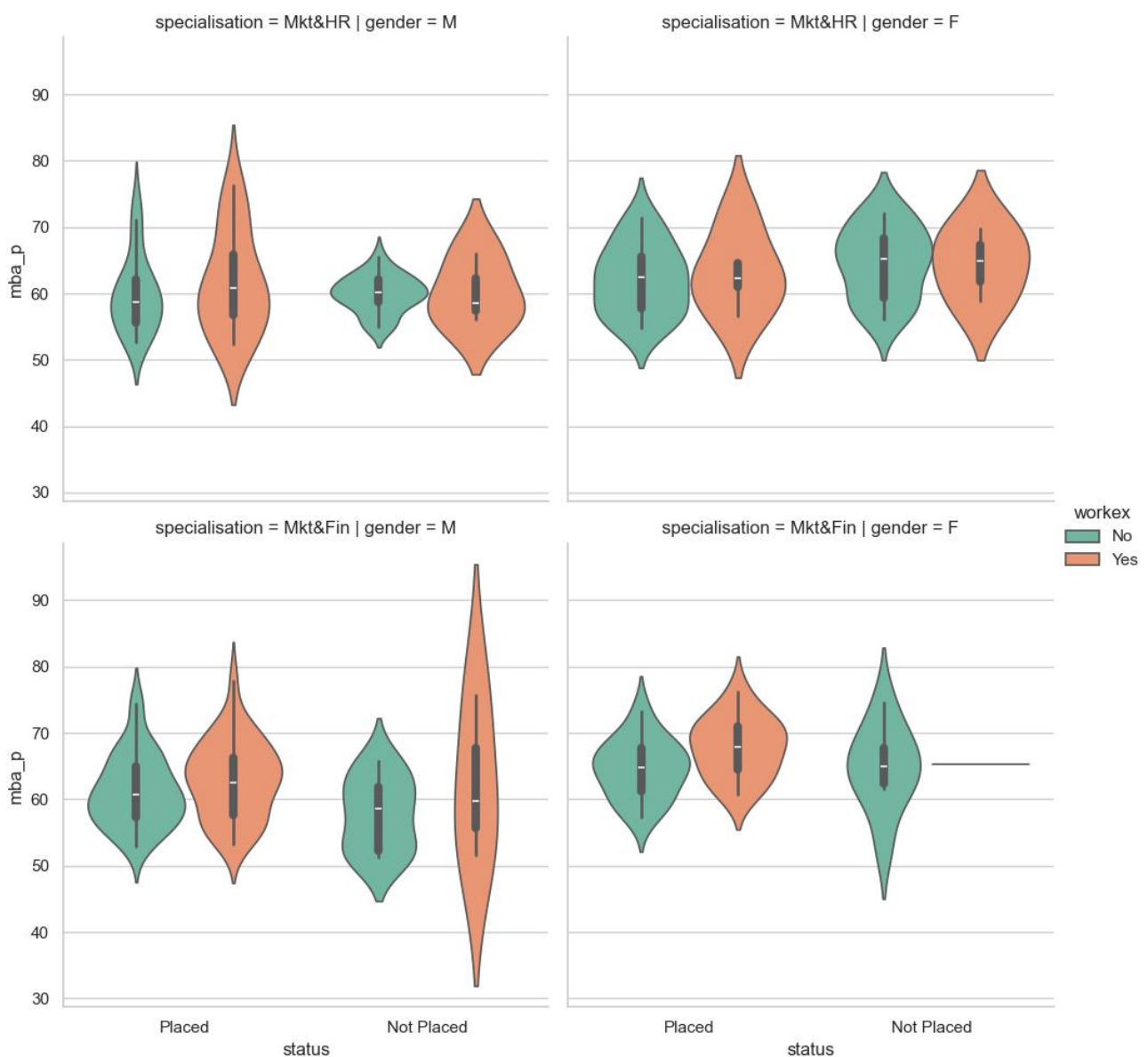
Factor Plot / Cat Plot

Factor Plot / CatPlot is a high-level categorical plot function in Seaborn.

It acts as a wrapper around other categorical plots (stripplot, swarmplot, boxplot, violinplot, barplot, countplot). It integrates with FacetGrid, so you can split plots across rows/columns by additional categorical variables.

```
sns.catplot(data=data, x='status', y='mba_p', col='gender', row='specialisation', hue='workex', kind='violin')
plt.show()
```

MBA Percent vs Placement Status



1. From the Plot, we can see For both genders, students with higher MBA% (closer to 70–80) are more often in the Placed group.
2. Across most panels, candidates with work experience ("Yes") show a slightly wider and higher violin body in the "Placed" category, suggesting that work experience gives a competitive edge for placements. Interestingly, in Mkt&Fin, Male, Not Placed, work experience candidates show the widest spread (30–90), indicating that work experience alone doesn't guarantee placement.
3. Mkt&HR students (both genders) show tighter distributions around the 60–70 MBA% range. Mkt&Fin has a more variable spread, especially for males, some placed at very high MBA% (90+), some at much lower (40–50).